F70LA Life Insurance Mathematics A (2025/26)
Tutorial 4

1. You are given the following table of values for ℓ_x and A_x as the age x increases, assuming an effective rate of interest of 4% per annum. Calculate:

x	ℓ_x	A_x
55	99 176.23	0.29962
56	99 072.49	0.31088
57	98 953.70	0.32250
58	98 817.64	0.33450
59	98 662.00	0.34685
60	98 484.31	0.35954

   (a) ${}_5p_{55}$.

   (b) $A^{1}_{55:\overline{5}|}$.

   (c) $A_{55:\overline{5}|}$.

   (d) The EPV of a whole life insurance policy issued to a life age exactly 60 years, with a sum insured of £50 000 and the death benefit payable at the end of year of death.

2. You are given that, at an interest rate of 4% per annum effective and on a certain mortality basis, $A_{61} = 0.37265$, $q_{59} = 0.001801$ and $q_{60} = 0.002058$. Hence calculate A_{59} and A_{60}.

3. Calculate A_{65} given

$$A_{45} = 0.24850, \qquad A^{1}_{45:\overline{20}|} = 0.03205 \qquad A_{45:\overline{20}|} = 0.46276.$$

4. You are given the following select life table.

x	$\ell_{[x]}$	$\ell_{[x]+1}$	$\ell_{[x]+2}$	$\ell_{[x]+3}$	ℓ_{x+4}	$x+4$
40	100 000	99 899	99 724	99 520	99 288	44
41	99 802	99 689	99 502	99 283	99 033	45
42	99 597	99 471	99 268	99 030	98 752	46
43	99 365	99 225	99 007	98 747	98 435	47
44	99 120	98 964	98 726	98 429	98 067	48

   Assuming an interest rate of 5% per annum effective, calculate:

   (a) $A^{1}_{[40]:\overline{2}|}$. (b) $A_{[41]+2:\overline{3}|}$.

5. (a) Describe in words the insurance benefits with present values given below.

(i)
$$Z_1 = \begin{cases} 200v^{T_x} & \text{if } T_x \leq 20, \\ 100v^{T_x} & \text{if } T_x > 20. \end{cases}$$

(ii)
$$Z_2 = \begin{cases} 0 & \text{if } T_x \leq 8, \\ 100v^{T_x} & \text{if } 8 < T_x \leq 20, \\ 100v^{20} & \text{if } T_x > 20. \end{cases}$$

(b) Write down in integral form the formula for the expected value of (i) Z_1 and (ii) Z_2.

(c) Derive expressions in terms of standard actuarial functions for the expected values of Z_1 and Z_2.

(d) Derive an expression in terms of standard actuarial functions for the covariance of Z_1 and Z_2.

6. Let Z_1 denote the present value of an n-year term insurance, issued to (x) and payable immediately upon death. Let Z_2 denote the present value of a whole life insurance, issued to the same life (x) and also payable immediately upon death.

(a) Express the covariance of Z_1 and Z_2 in actuarial functions, simplified as much as possible.

Let Z_3 denote the present value of a whole life insurance, payable immediately upon death but issued to a different life (y). Assume that the future lifetimes of (x) and (y) can be modelled as independent random variables.

(b) State the covariance of Z_1 and Z_3 and explain mathematically, using words, your answer.

Numerical solutions: 1(a) 0.99302; 1(b) 0.00617; 1(c) 0.82236; 1(d) £17 997.00. Q2. $A_{59} = 0.34684$ and $A_{60} = 0.35956$. Q3. 0.50254. Q4(a) 0.002549; 4(b) 0.86414. Q6(a) ${}^2\bar{A}^1_{x:\overline{n}|} - \bar{A}^1_{x:\overline{n}|}\,\bar{A}_x$; Q6(b) 0.